

$e^{j\frac{2\pi}{P}kt}$ is defined for all $k \in \mathbb{Z}$

$$x(t) = \underbrace{\frac{x(t) + x(-t)}{2}}_{\text{even}} + \underbrace{\frac{x(t) - x(-t)}{2}}_{\text{odd}}$$

$$\lim_{N \rightarrow \infty} \sum_{k=-N}^N x_k e^{j\frac{2\pi}{P}kt} = x(t)$$

order N approximation

G_s
even function
sin
odd

Fourier Series:

- A periodic signal, with period P , satisfies

$$x(t) = x(t + nP) \text{ for all integers } n.$$

Examples: sine wave, square wave, triangle wave, ...

We consider the vector space of all periodic signals with period P .

- The Fourier series (FS) says that a P -periodic signal $x(t)$ can be written as an infinite sum of complex exponentials:

$$x(t) = \sum_{k=-\infty}^{\infty} x_k e^{j\frac{2\pi}{P}kt}, \quad x_k \triangleq \frac{1}{P} \int_0^P x(t) e^{-j\frac{2\pi}{P}kt} dt,$$

Complex sinusoid of freq. $\frac{2\pi}{P}k \rightarrow P$ -periodic

where $\{x_k\}_{k \in \mathbb{Z}}$ are known as the FS coefficients.

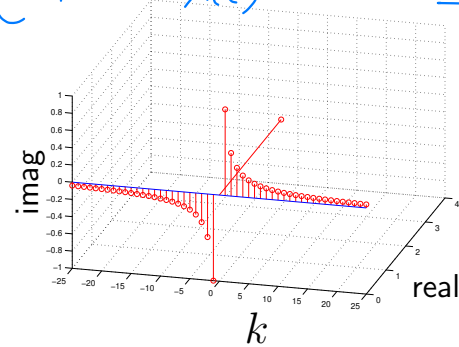
$e^{j\frac{2\pi}{P}k(t+P)} = e^{j\frac{2\pi}{P}kt} e^{j2\pi k}$

$e^{j\frac{2\pi}{P}kt} \cdot x(t)$

- Example — sawtooth wave:

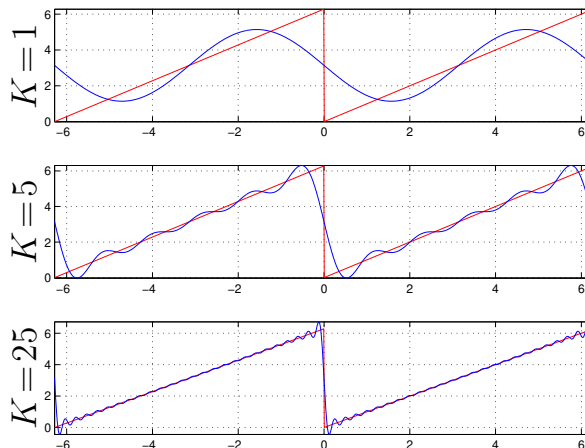
$$x(t) = t \text{ for } t \in [0, P), P = 2\pi$$

$$x_k = \frac{j}{k} |_{k \neq 0}, \quad x_0 = \frac{P}{2}$$



K -truncated FS

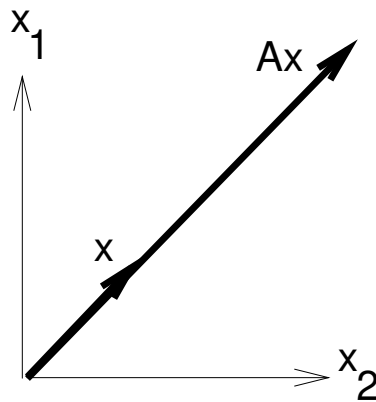
$$\sum_{k=-K}^K x_k e^{j\frac{2\pi}{P}kt}$$



Eigenfunctions of DT LTI Systems:

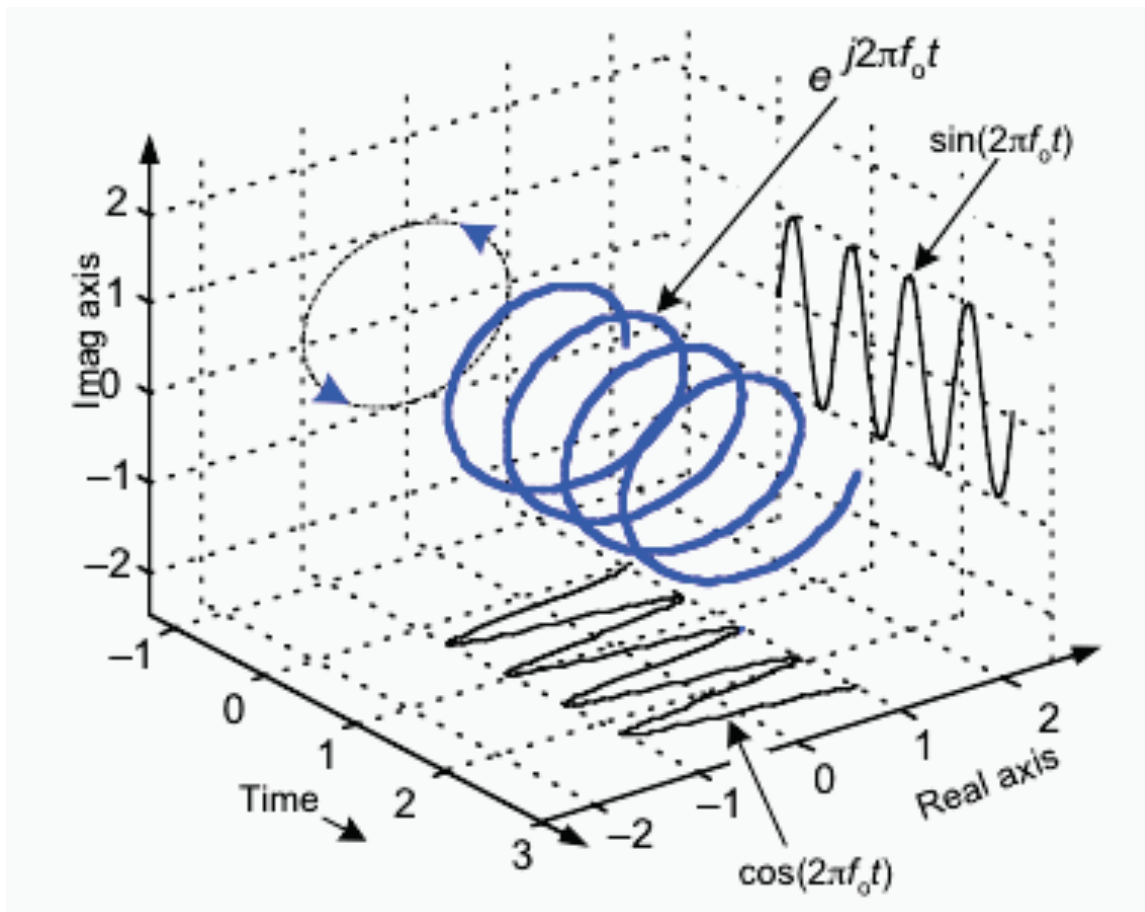
- An analogy:

$$A\mathbf{x} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



- LTI System:

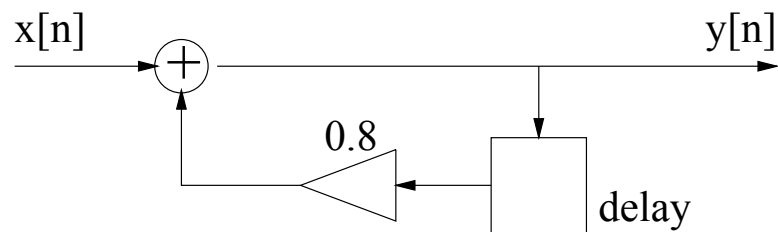
$$\{e^{j\omega n}\}_{n \in \mathbb{Z}} \longrightarrow \boxed{\mathcal{H}} \longrightarrow \lambda_{\omega} \cdot \{e^{j\omega n}\}_{n \in \mathbb{Z}}$$



IIR Example:

Consider a recursive LTI system:

$$y[n] = x[n] + 0.8y[n - 1]$$



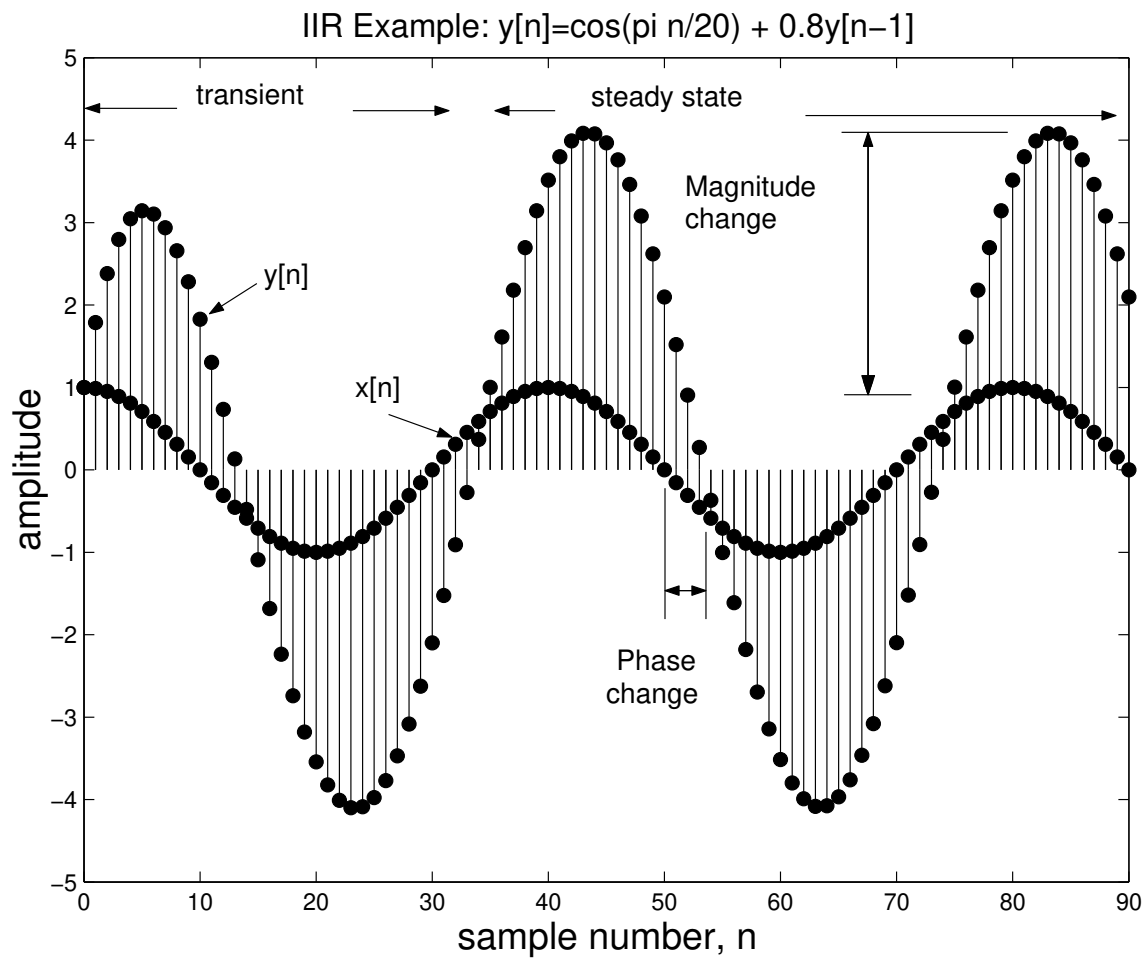
$$x[n] = \cos(\pi n/20)u[n]$$

$$|H(e^{j\omega})| = 4.093 \quad \text{for } \omega = \pi/20$$

$$\angle H(e^{j\omega}) = -0.538 \text{ radians} \quad \text{for } \omega = \pi/20$$

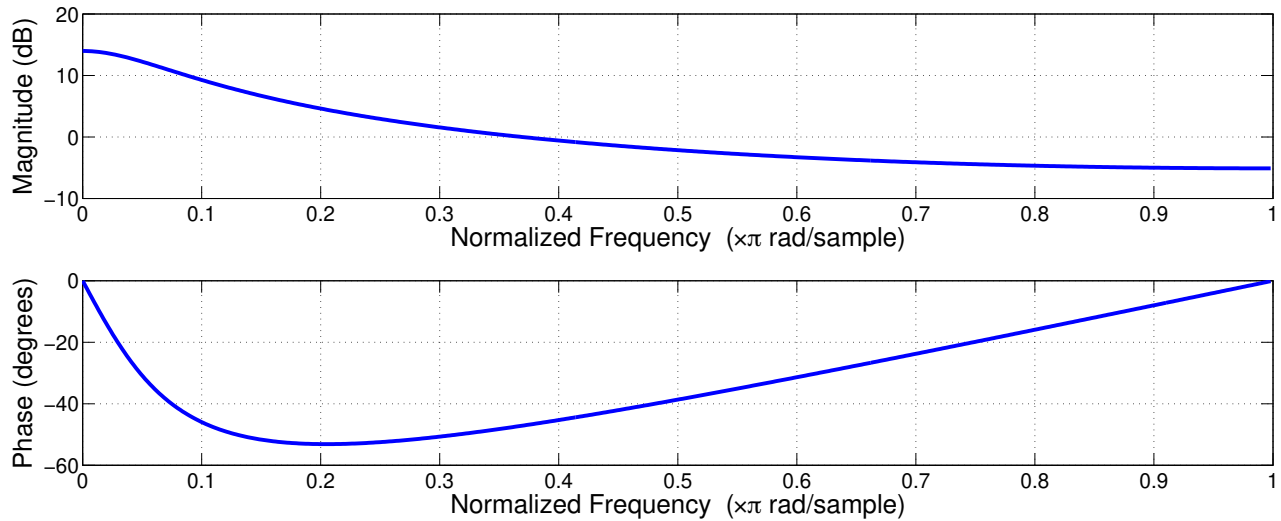
$$y_{ss}[n] = 4.093 \cos(\pi n/20 - 0.538)$$

IIR Example, continued:



IIR Example, continued:

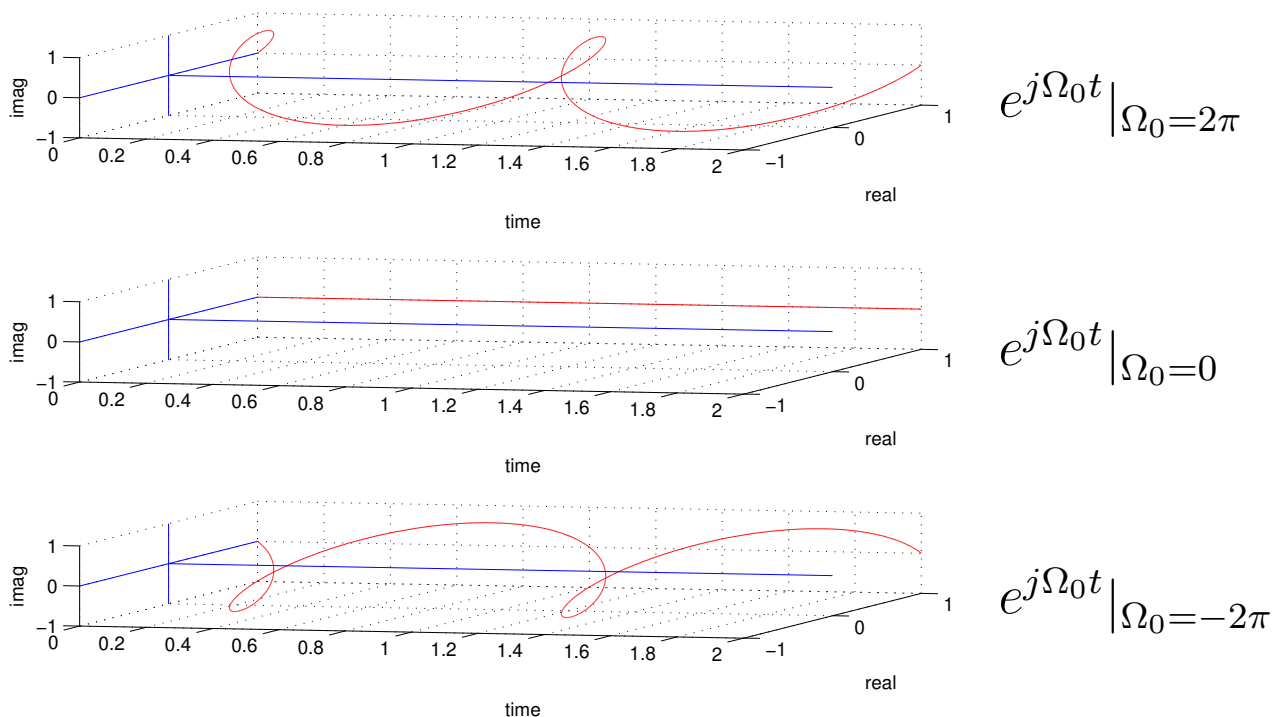
Frequency response plots (magnitude and phase)



```
>> freqz([1],[1 -0.8])
```

Negative frequencies:

- The CTFT allows negative frequencies. What are they?
- Recall that a “pure” tone at Ω_0 radians/sec can be written as $e^{j\Omega_0 t} \xleftrightarrow{\text{CTFT}} 2\pi\delta(\Omega - \Omega_0)$.
- The signal $e^{j\Omega_0 t}$ is a corkscrew: for $\Omega_0 > 0$, it spins counter-clockwise and, for $\Omega_0 < 0$, it spins clockwise.



- In essence, the CTFT implies that any signal can be constructed as a sum of (possibly infinitely many) scaled corkscrews, each with a distinct frequency. The sign of the frequency simply denotes the direction of spin.

- What if the signal is a real-valued sinusoid?

$$\begin{aligned}\cos(\Omega_0 t + \theta) &= \frac{1}{2} \left(e^{j(\Omega_0 t + \theta)} + e^{-j(\Omega_0 t + \theta)} \right) \\ &= \frac{e^{j\theta}}{2} e^{j\Omega_0 t} + \frac{e^{-j\theta}}{2} e^{-j\Omega_0 t}\end{aligned}$$

Thus it contains both positive and negative frequencies.

- How do we generate/represent complex signals in hardware?

$$e^{j(\Omega_0 t + \theta)} = \cos(\Omega_0 t + \theta) + j \sin(\Omega_0 t + \theta)$$

Using two real-valued wires/buses.

FIR Example:

$$y[n] = 0.5x[n] + x[n - 1] + 0.5x[n - 2]$$

This is a second-order FIR filter, and hence is a linear time-invariant system. We have that

$$h[n] = 0.5\delta[n] + \delta[n - 1] + 0.5\delta[n - 2]$$

To find $H(e^{j\omega})$, we start with the definition

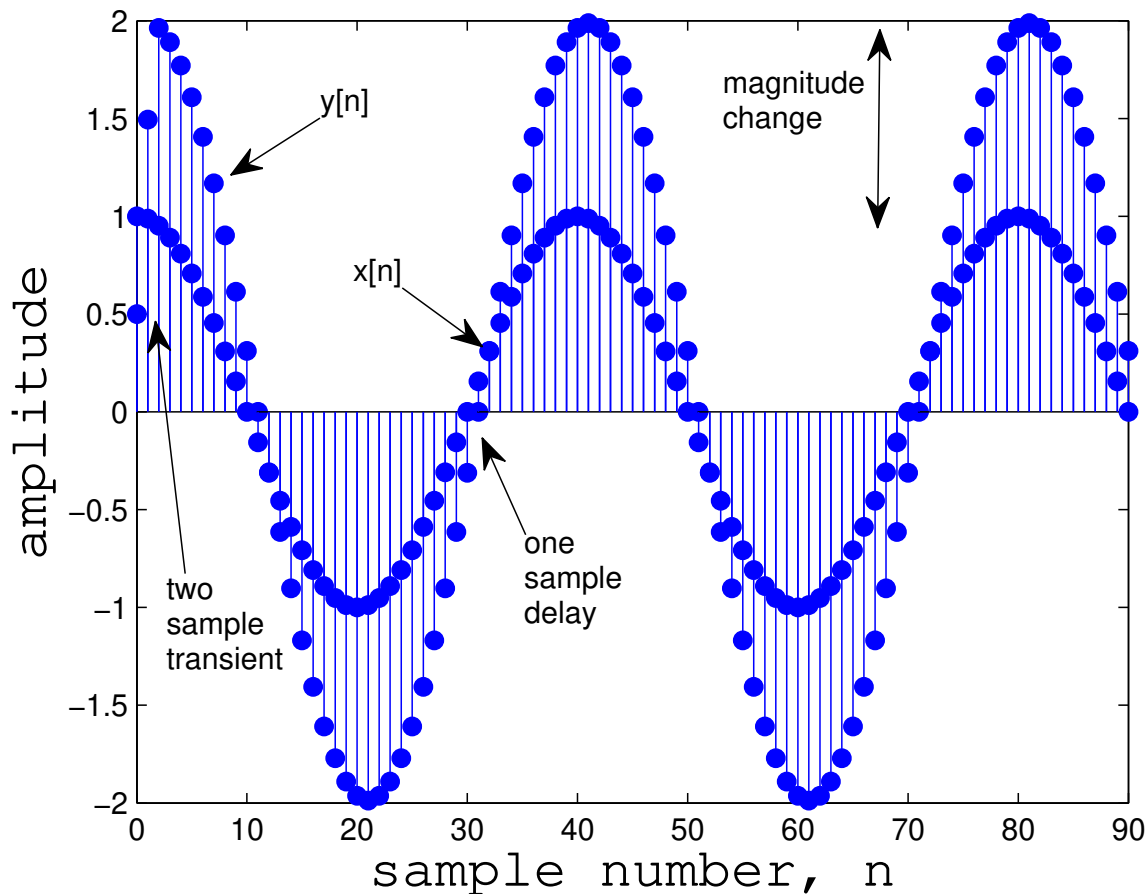
$$\begin{aligned} H(e^{j\omega}) &= \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \\ &= \frac{1}{2}e^0 + 1e^{-j\omega} + \frac{1}{2}e^{-j\omega 2} \\ &= e^{-j\omega} \left\{ \frac{1}{2}e^{j\omega} + 1 + \frac{1}{2}e^{-j\omega} \right\} \\ &= e^{-j\omega} \{1 + \cos(\omega)\} \end{aligned}$$

Thus, $|H(e^{j\omega})| = 1 + \cos(\omega)$ and $\angle H(e^{j\omega}) = -\omega$.

FIR Example, continued:

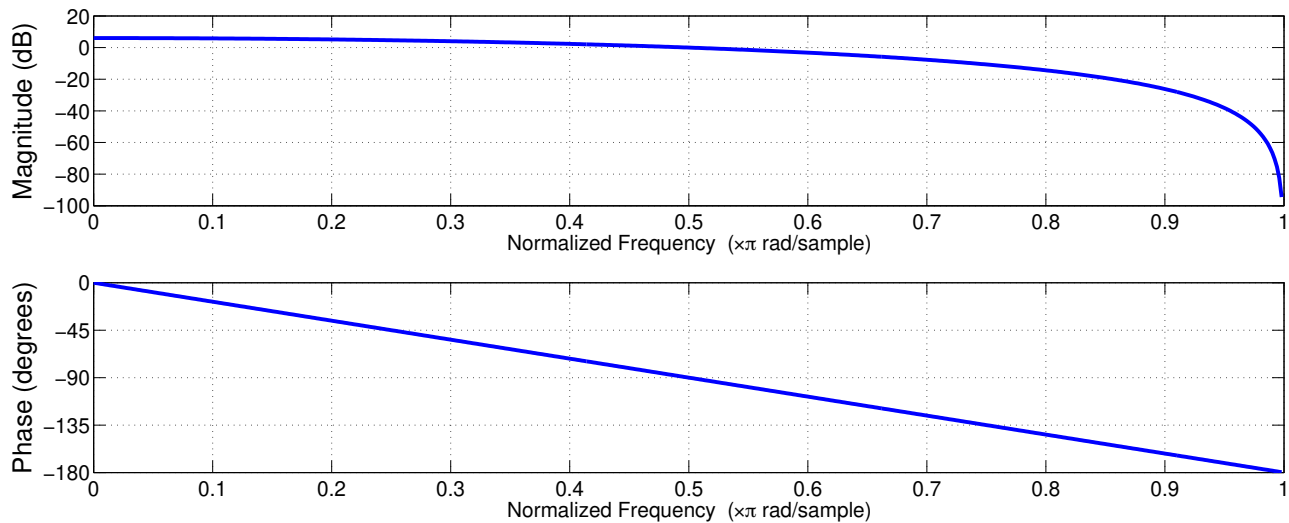
For $x[n] = \cos(\pi n/20)u[n]$ we have $A = 1$, $\omega = \pi/20$ and $\phi = 0$. Thus,

$$\begin{aligned}y_{ss}[n] &= A |H(e^{j\omega})| \cos(\omega n + \phi + \angle H(e^{j\omega})) \\&= 1 \cdot |H(e^{j\pi/20})| \cos(\pi n/20 + 0 + \angle H(e^{j\pi/20})) \\&= 1.988 \cos(\pi n/20 - \pi/20)\end{aligned}$$



FIR Example, continued:

Frequency response plots (magnitude and phase)



```
>> b = [0.5, 1, 0.5];  
>> a=1;  
>> freqz(b,a);
```

Eigensignals of CT-LTI systems:

- Consider complex *sinusoidal* input $x(t) = e^{j\Omega_0 t}$, $\Omega_0 \in \mathbb{R}$.

For LTI system $\mathcal{H}_c$, we have

$$\begin{aligned}\mathcal{H}_c\{x(t)\} &= \int h(\tau)x(t-\tau)d\tau = \int h(\tau)e^{j\Omega_0(t-\tau)}d\tau \\ &= e^{j\Omega_0 t} \underbrace{\int h(\tau)e^{-j\Omega_0\tau}d\tau}_{\triangleq H_c(j\Omega_0)} = x(t) \underbrace{H_c(j\Omega_0)}_{\text{scaling factor}},\end{aligned}$$

using Laplace-transform notation (more on this later...).

The output is simply a scaled version of the input!

- A signal which passes through system $\mathcal{H}_c$ with only a gain/phase change is an “*eigensignal*” of $\mathcal{H}_c$, and the corresponding scaling factor is its “*eigenvalue*.”

Sinusoids are eigensignals of LTI systems.

- This property yields an important implication for any *P-periodic signal* $x(t)$. Notice that if we expand such a signal via its *Fourier series* (FS) as

$$x(t) = \sum_{k=-\infty}^{\infty} x_k e^{j\frac{2\pi}{P}kt}$$

then the linearity of $\mathcal{H}_c$ implies

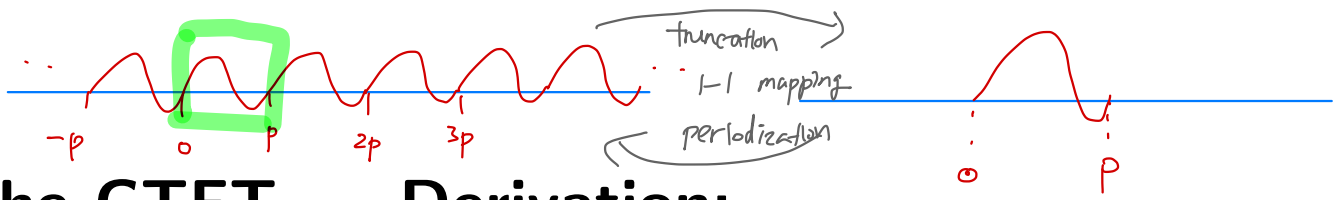
$$\mathcal{H}_c\{x(t)\} = \sum_{k=-\infty}^{\infty} \underbrace{x_k H_c(j\frac{2\pi k}{P})}_{y_k} \overbrace{e^{j\frac{2\pi}{P}kt}}^{h_k}.$$

- Thus, if we represent the input signal by $\{x_k\}_{k \in \mathbb{Z}}$, the output signal by $\{y_k\}_{k \in \mathbb{Z}}$, and the system by $\{h_k\}_{k \in \mathbb{Z}}$, then the operation of passing the signal through the system is simply multiplication of the corresponding coefficients: $y_k = h_k x_k$!

For an LTI system with a periodic input, the output can be computed algebraically.

- Note, however, that the example above applies only to *periodic* signals. Can we extend it to non-periodic signals?

Yes, through the continuous-time Fourier transform (CTFT)...

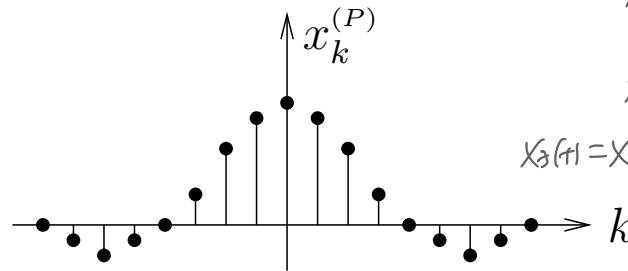
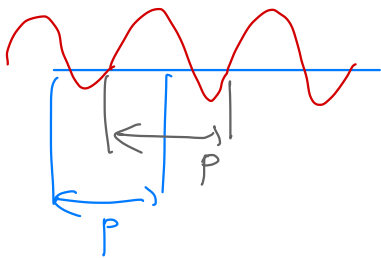


The CTFT — Derivation:

- We now derive the *continuous-time Fourier transform* (CTFT) as the $P \rightarrow \infty$ limit of the Fourier series (FS).
- Consider an arbitrary P -periodic signal $x^{(P)}(t)$ with $x(t)$ P -periodic version of corresponding FS coefficients $\{x_k^{(P)}\}_{k \in \mathbb{Z}}$, where

$$x_k^{(P)} = \frac{1}{P} \int_0^P \underbrace{x^{(P)}(t)}_{P\text{-periodic}} \underbrace{e^{-j\frac{2\pi}{P}kt}}_{P\text{-periodic}} dt = \frac{1}{P} \int_{-P/2}^{P/2} x^{(P)}(t) e^{-j\frac{2\pi}{P}kt} dt.$$

- The collection $\{x_k^{(P)}\}_{k \in \mathbb{Z}}$ constitutes the FS “spectrum”:



$x_1(t)$ P -periodic

$$x_1(t + mP) = x_1(t)$$

$x_2(t)$ P -periodic

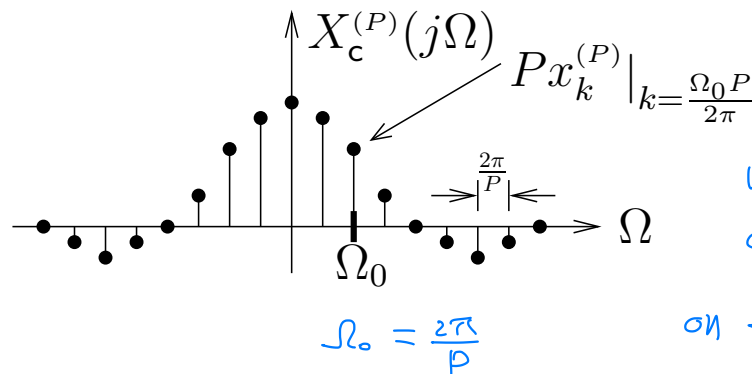
$$x_3(t) = x_1(t) x_2(t)$$

$$\begin{aligned} x_3(t + mP) &= x_1(t + mP) x_2(t + mP) \\ &= x_1(t) x_2(t) \\ &= x_3(t) \end{aligned}$$

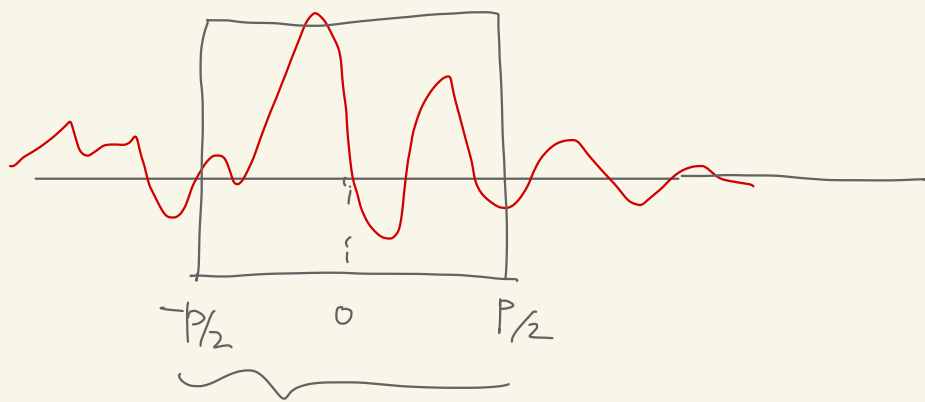
- Now define the “frequency” $\Omega \triangleq \frac{2\pi k}{P}$ (in rad/sec) and, from it, the (scaled) spectrum

$$X_c^{(P)}(j\Omega) \Big|_{\Omega = \frac{2\pi k}{P}} \triangleq P x_k^{(P)} = \int_{-P/2}^{P/2} x^{(P)}(t) e^{-j\Omega t} dt. \Big|_{\Omega = \frac{2\pi k}{P}}$$

- We can visualize the spectrum $X_c^{(P)}(j\Omega)$ as follows:



We know samples of $X_c^{(P)}(j\Omega)$ on the grid.

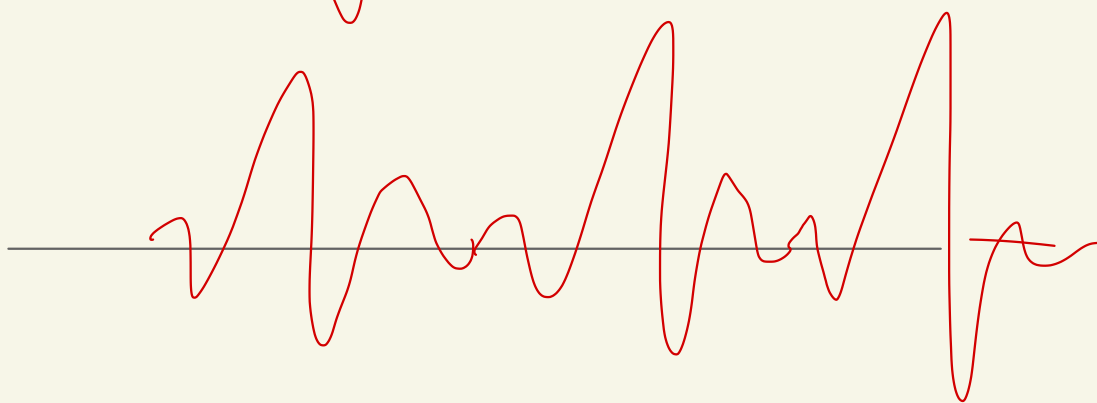


truncation

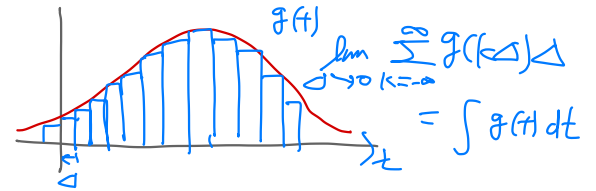


support
of length P

periodization



Riemann integral



Synthesis
FS representation

- Using FS, we can relate $X_c^{(P)}(j\Omega)$ to $x^{(P)}(t)$ as follows:

$$x^{(P)}(t) = \sum_{k=-\infty}^{\infty} x_k^{(P)} e^{j\frac{2\pi}{P}tk} = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X_c^{(P)}(j\Omega) e^{j\Omega t} \frac{2\pi}{P} \Big|_{\Omega = \frac{2\pi k}{P}}$$

- Key idea: as P increases, the # of spectral lines per Hz in $X_c^{(P)}(j\Omega)$ increases, so that $X_c^{(P)}(j\Omega)$ approaches a continuum! In this case, Riemann sum arguments allow us to write the infinite sum as an integral. So, for $x(t) \triangleq \lim_{P \rightarrow \infty} x^{(P)}(t)$, which is no longer periodic, and $X_c(j\Omega) \triangleq \lim_{P \rightarrow \infty} X_c^{(P)}(j\Omega)$, we get

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) e^{j\Omega t} d\Omega$$

$P X_k^{(P)} \longrightarrow X_c(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt, \Big|_{\Omega = \frac{2\pi k}{P}}$

known as the CTFT.

$e^{j\Omega t} \cdot x(t)$

- The convolution operation $y(t) = \int h(\tau) x(t - \tau) d\tau$ becomes algebraic under the CTFT:

$$\begin{aligned} Y_c(j\Omega) &= \int \left[\int h(\tau) x(t - \tau) d\tau \right] e^{-j\Omega t} dt \\ &= \int \int h(\tau) e^{-j\Omega \tau} x(t - \tau) e^{-j\Omega(t - \tau)} dt d\tau \\ &= \int h(\tau) e^{-j\Omega \tau} d\tau \int x(\rho) e^{-j\Omega \rho} d\rho \\ &= H_c(j\Omega) X_c(j\Omega). \end{aligned}$$

Fourier Transform and Its Properties

- **Synthesis (inverse Fourier transform)** *scalar-valued weight*

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) e^{j\Omega t} d\Omega$$

- **Analysis (Fourier transform)**

$$X_c(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt = e^{j\Omega t} \cdot x(t)$$

measures the similarity between $X(t)$ and $e^{j\Omega t}$

- Pairs of representations in the time-domain and in the frequency domain provide useful properties (similar to DFT and Fourier series)
- Derived from the definitions of the synthesis/analysis operators and from the properties of complex sinusoids
- Handy tools for understanding key results and easy calculation

Linearity

$$x(t) \longleftrightarrow X_c(j\Omega)$$

$X_c(j\Omega)$ is CTFT of $x(t)$

Ω : freq
(rad/sec)

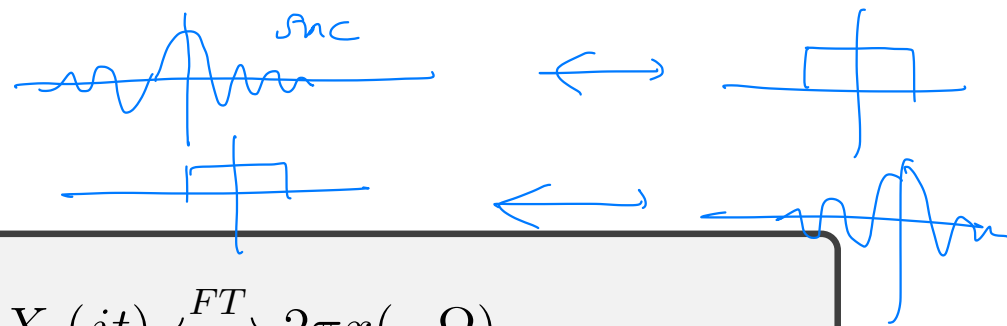
$$ax(t) + by(t) \xrightarrow{FT} aX_c(j\Omega) + bY_c(j\Omega)$$

Proof: FT pair is one-to-one. It suffices to show the implication in any one direction. For example,

$$\underbrace{\int_{-\infty}^{\infty} [ax(t) + by(t)] e^{-j\Omega t} dt}_{\text{FT of } ax(t) + by(t)} = a \underbrace{\int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt}_{X_c(j\Omega)} + b \underbrace{\int_{-\infty}^{\infty} y(t) e^{-j\Omega t} dt}_{Y_c(j\Omega)}.$$

Alternatively, one can show

$$\begin{aligned} & \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} [aX_c(j\Omega) + bY_c(j\Omega)] e^{j\Omega t} d\Omega}_{\text{IFT of } aX_c(j\Omega) + bY_c(j\Omega)} \\ &= a \cdot \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) e^{j\Omega t} d\Omega}_{x(t)} + b \cdot \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} Y_c(j\Omega) e^{j\Omega t} d\Omega}_{y(t)}. \end{aligned}$$



Duality

$$x(t) \xleftrightarrow{FT} X_c(j\Omega) \implies X_c(jt) \xleftrightarrow{FT} 2\pi x(-\Omega)$$

Proof: Let $y(t) = X_c(jt)$. Then we compute the FT of $y(t)$ at $\Omega = \theta$ by

$$\begin{aligned} Y_c(j\theta) &= \int_{-\infty}^{\infty} y(t) e^{-j\theta t} dt = \int_{-\infty}^{\infty} X_c(jt) e^{-j\theta t} dt \\ &= 2\pi \cdot \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) e^{j(-\theta)\Omega} d\Omega}_{\text{IFT of } X_c(j\Omega) \text{ at } t = -\theta} = 2\pi x(-\theta). \end{aligned}$$

Alternatively, let $Z_c(j\Omega) = 2\pi x(-\Omega)$. Then the IFT of $Z_c(j\Omega)$ at $t = \theta$ is

$$\begin{aligned} z(\theta) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} Z_c(j\Omega) e^{j\Omega\theta} d\Omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi x(-\Omega) e^{j\Omega\theta} d\Omega \\ &= \underbrace{\int_{-\infty}^{\infty} x(t) e^{-j\theta t} dt}_{\text{FT of } x(t) \text{ at } \Omega = \theta} = X_c(j\theta) \end{aligned}$$

Time-Shift / Freq-Shift

$$x(t - \tau) \xleftrightarrow{FT} e^{-j\Omega\tau} X_c(j\Omega)$$

Proof:

$$\int_{-\infty}^{\infty} x(t - \tau) e^{-j\Omega t} dt = \int_{-\infty}^{\infty} x(\lambda) e^{-j\Omega(\lambda + \tau)} d\lambda = e^{-j\Omega\tau} \int_{-\infty}^{\infty} x(\lambda) e^{-j\Omega\lambda} d\lambda$$

$$e^{j\nu t} x(t) \xleftrightarrow{FT} X_c(j(\Omega - \nu))$$

modulation

Proof:

$$\begin{aligned} \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j(\Omega - \nu)) e^{j\Omega t} d\Omega &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\lambda) e^{j(\lambda + \nu)t} d\lambda \\ &= e^{j\nu t} \cdot \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\lambda) e^{j\lambda t} d\lambda \end{aligned}$$

Conjugation and Symmetry

$$x(t)^* \xleftrightarrow{FT} X_c(-j\Omega)^*$$

Proof:

$$\int_{-\infty}^{\infty} x(t)^* e^{-j\Omega t} dt = \int_{-\infty}^{\infty} x(t)^* (e^{-j(-\Omega)t})^* dt = \left(\int_{-\infty}^{\infty} x(t) e^{-j(-\Omega)t} dt \right)^*$$

$$\text{\textit{x(t) is real-valued}} \quad x(t)^* = x(t) \implies X_c(j\Omega) = X_c(-j\Omega)^*$$

Proof: Linearity implies

$$\text{Re}(X_c(j\Omega)) + j\text{Im}(X_c(j\Omega)) = \text{Re}(X_c(-j\Omega)) - j\text{Im}(X_c(-j\Omega)).$$

Therefore,

$$|X_c(j\Omega)| = |X_c(-j\Omega)|, \quad \angle X_c(j\Omega) = -\angle X_c(-j\Omega).$$

Differentiation/Integration

very important
to solve PDE

$$\frac{dx(t)}{dt} \xleftrightarrow{FT} j\Omega X_c(j\Omega)$$

Proof:

$$\begin{aligned} \frac{d}{dt} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) e^{j\Omega t} d\Omega \right) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) \frac{d}{dt} e^{j\Omega t} d\Omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) j\Omega e^{j\Omega t} d\Omega \end{aligned}$$

$$\int_{-\infty}^t x(\lambda) d\lambda \xleftrightarrow{FT} \frac{X_c(j\Omega)}{j\Omega} + \pi X_c(j0) \delta(\Omega)$$

We indeed compute the FT of $x(t) * u(t)$. We will revisit the proof after the convolution property and the FT of $u(t)$.

Time-Scaling (Compression/Stretching)

$$x(at) \xleftrightarrow{FT} \frac{X_c(j\Omega/a)}{a}$$

Proof:

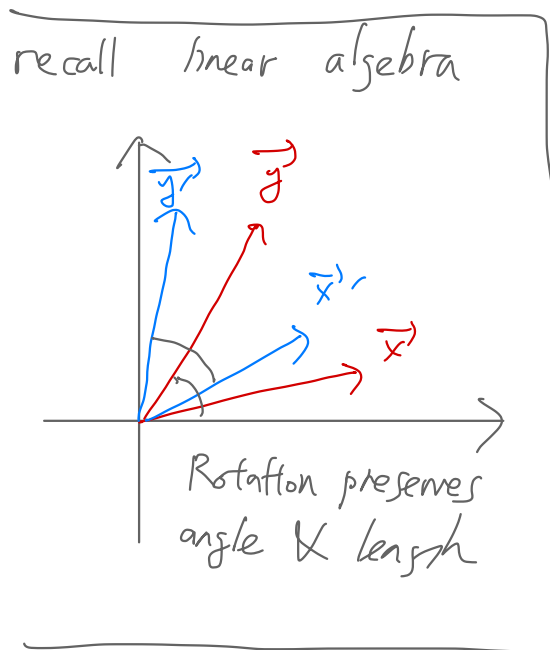
$$\int_{-\infty}^{\infty} x(at)e^{-j\Omega t} dt = \int_{-\infty}^{\infty} x(\lambda)e^{-j(\Omega/a)\lambda} \frac{d\lambda}{a} = \frac{1}{a} \int_{-\infty}^{\infty} x(\lambda)e^{-j(\Omega/a)\lambda} d\lambda$$

Parseval's Identity

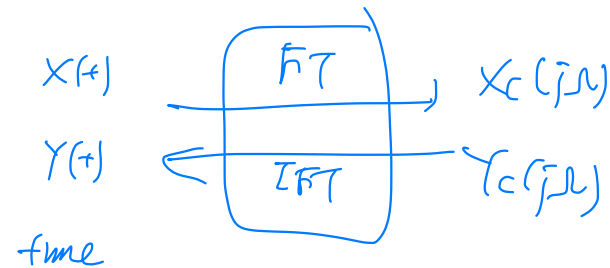
$$Y_c(j\Omega) \cdot X_c(j\Omega)$$

$$y(t) \cdot x(t) = \int_{-\infty}^{\infty} x(t)y(t)^* dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega)Y_c(j\Omega)^* d\Omega$$

Proof:



$$\begin{aligned} \int_{-\infty}^{\infty} x(t)y(t)^* dt &= \int_{-\infty}^{\infty} \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega)e^{j\Omega t} d\Omega y(t)^* dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) \int_{-\infty}^{\infty} y(t)^* e^{-j(-\Omega)t} dt d\Omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega)Y_c(j\Omega)^* d\Omega \end{aligned}$$



$$x(t) = y(t) \quad \|x(t)\|^2 = \|X_c(j\Omega)\|^2$$

FT preserves the length and the dot product.

$$X(t) = \frac{1}{2\pi} \int X_c(j\omega) e^{j\omega t} d\omega$$

Convolution!!! $h(t) * x(t) = \frac{1}{2\pi} \int X_c(j\omega) (h(t) * e^{j\omega t}) d\omega$

$$\int_{-\infty}^{\infty} x(t - \lambda) y(\lambda) d\lambda \xleftrightarrow{FT} X_c(j\Omega) Y_c(j\Omega) = \frac{1}{2\pi} \int X_c(j\omega) H_c(j\omega) e^{j\omega t} d\omega$$

Proof: Time-shift property implies

$$x(t - \lambda) \xleftrightarrow{FT} e^{-j\lambda\Omega} X_c(j\Omega).$$

Then

$$\begin{aligned} & \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) Y_c(j\Omega) e^{j\Omega t} d\Omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) \left(\int_{-\infty}^{\infty} y(\lambda) e^{-j\Omega\lambda} d\lambda \right) e^{j\Omega t} d\Omega \\ &= \int_{-\infty}^{\infty} y(\lambda) \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-j\lambda\Omega} X_c(j\Omega) e^{j\Omega t} d\Omega \right) d\lambda \\ &= \int_{-\infty}^{\infty} y(\lambda) x(t - \lambda) d\lambda. \end{aligned}$$

$$e^{j\omega t} \rightarrow \boxed{h(t)} \rightarrow$$

$$y(t) = h(t) * e^{j\omega t}$$

$$= \int h(\lambda) e^{j\omega(t-\lambda)} d\lambda$$

$$= e^{j\omega t} \underbrace{\int h(\lambda) e^{-j\omega\lambda} d\lambda}_{H_c(j\omega)}$$

$e^{j\omega t}$ is an
eigenfunction
with eigenvalue
 $H_c(j\omega)$

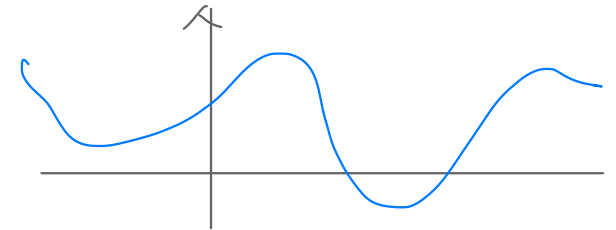
$$\frac{y(t)}{x(t)} = \frac{H_c(j\omega) e^{j\omega t}}{e^{j\omega t}}$$

Hadamard Product (Pointwise Multiplication)

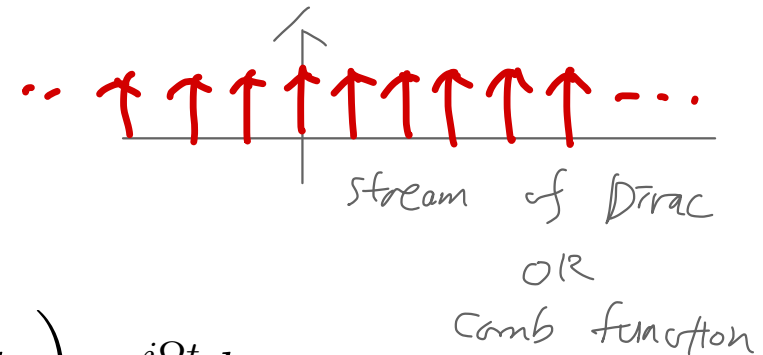
$$x(t)y(t) \xleftrightarrow{FT} \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j(\Omega - \mu))Y_c(j\mu)d\mu$$

Proof: Freq-shift property implies

$$x(t)e^{j\mu t} \xleftrightarrow{FT} X_c(j(\Omega - \mu)).$$

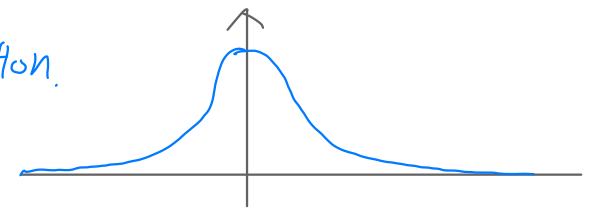


Then



$$\begin{aligned} & \int_{-\infty}^{\infty} x(t)y(t)e^{-j\Omega t}dt \\ &= \int_{-\infty}^{\infty} x(t) \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} Y_c(j\mu)e^{j\mu t}d\mu \right) e^{-j\Omega t}dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} Y_c(j\mu) \left(\int_{-\infty}^{\infty} x(t)e^{j\mu t}e^{-j\Omega t}dt \right) d\mu \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} Y_c(j\mu)X_c(j(\Omega - \mu))d\mu \end{aligned}$$

Normal / Gaussian random variable or distribution.



Application: FT of Gaussian Function

probability density function (pdf) of Gaussian. $\longleftrightarrow$ $x(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{t^2}{2\sigma^2}}$ $\xleftrightarrow{FT}$ $e^{-\frac{\sigma^2\Omega^2}{2}}$

with zero mean & variance σ^2

$x(t) \geq 0$
 $\int x(t) dt = 1$
 $X_c(j0) = 1$

Proof:

$$x(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{t^2}{2\sigma^2}} \implies \frac{dx(t)}{dt} = -\frac{t}{\sigma^2} x(t).$$

$$\int \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{t^2}{2\sigma^2}} e^{-j\omega t} dt$$

Recall the derivative properties:

$$\frac{dx(t)}{dt} \xleftrightarrow{FT} j\Omega X_c(j\Omega), \quad -jtx(t) \xleftrightarrow{FT} \frac{dX_c(j\Omega)}{d\Omega}$$

Then we have

$$j\Omega X_c(j\Omega) = \frac{1}{j\sigma^2} \frac{dX_c(j\Omega)}{d\Omega}.$$

By rearranging terms together with the derivative of log, we obtain

$$\frac{d}{d\Omega} [\ln X_c(j\Omega)] = \frac{1}{X_c(j\Omega)} \cdot \frac{dX_c(j\Omega)}{d\Omega} = -\sigma^2\Omega.$$

Application: FT of Gaussian Function

By taking the integral with respect to Ω , we get

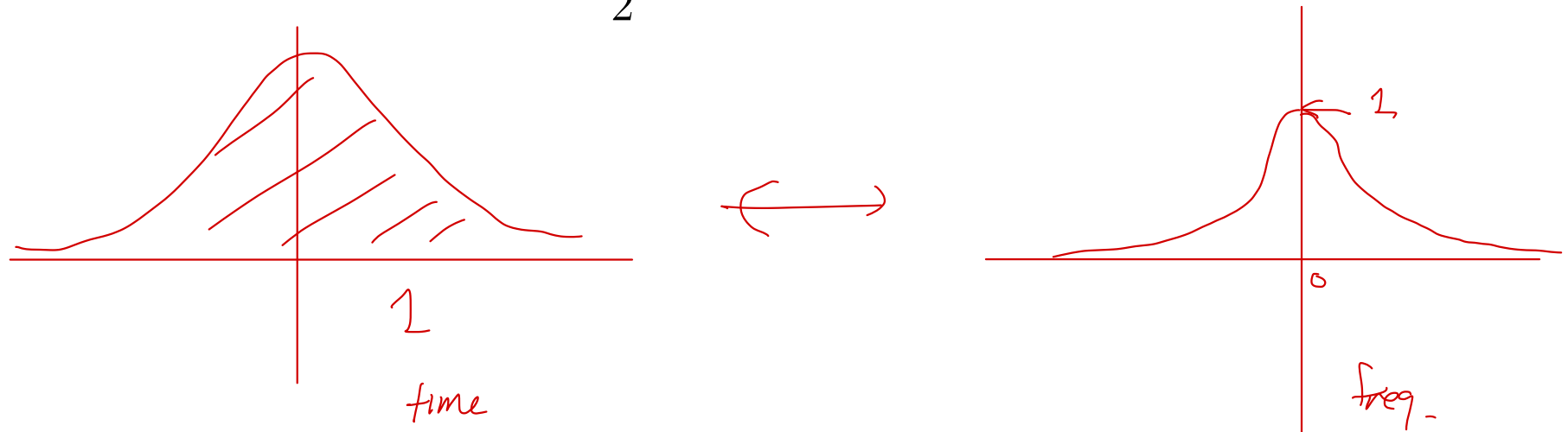
$$\ln X_c(j\Omega) - \ln X_c(j0) = -\frac{\sigma^2 \Omega^2}{2}.$$

On the other hand,

$$X_c(j0) = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{t^2}{2\sigma^2}} dt = 1.$$

Therefore,

$$\ln X_c(j\Omega) = -\frac{\sigma^2 \Omega^2}{2} \implies X_c(j\Omega) = e^{-\frac{\sigma^2 \Omega^2}{2}}.$$



bigger σ makes $x(t)$ wider $X_c(j\omega)$ narrower